



HOSPITALIA

HEALTH & WELLNESS

Patient Experience Analytics Project

NPS-Driven Healthcare Performance & Profitability Study

Team Member	Role & Contribution
Amr Zaki	Data Cleaning & Modeling, SQL, Power BI
Omnia Magdy	Tableau Dashboards & Presentation
Rawan Mahmoud	Data Analysis, Excel Dashboard, Python, Documentation
Fatma Ali	Excel Statistics & Descriptive Analysis, power Bi
Mostafa Samir	Helped in insights



1. Project Planning & Management

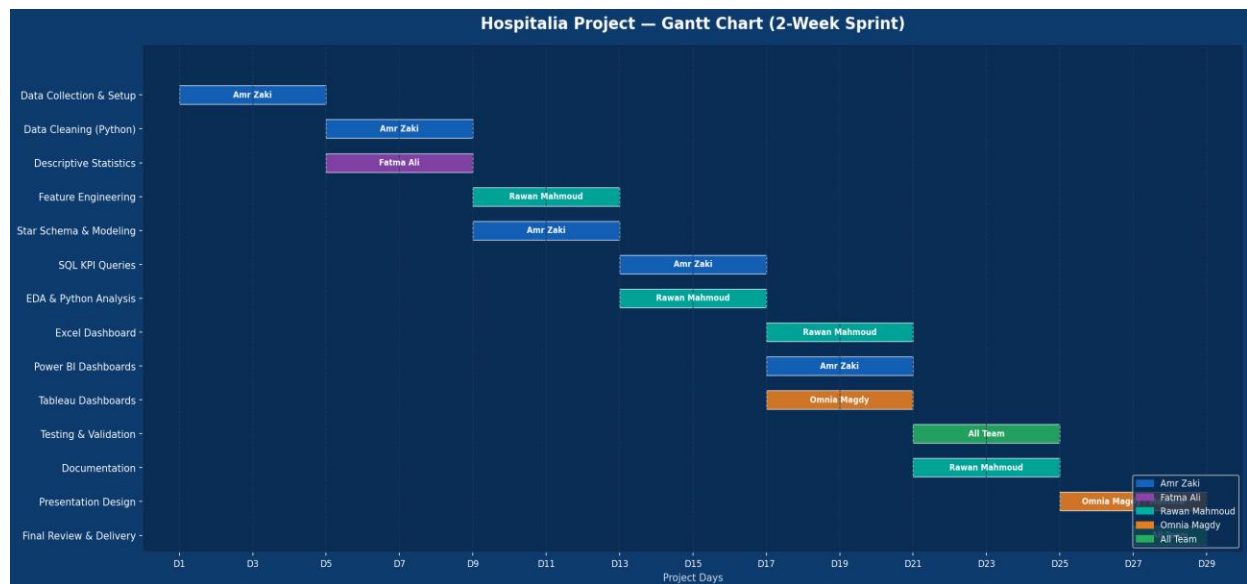
1.1 Project Proposal & Objectives

Hospitalia Health & Wellness operates a multi-branch hospital network across five regions (Central, North, South, East, West). The organization identified a declining Net Promoter Score (NPS) and its correlation with reduced profitability, prompting a full data analytics study.

- Identify the root causes of NPS decline across departments and branches.
- Quantify the statistical correlation between NPS and hospital profitability.
- Profile patient segments (Insurance, Regular, VIP) and satisfaction patterns.
- Deliver interactive dashboards for real-time operational decision-making.
- Produce a reproducible Python analytical pipeline with full documentation.

1.2 Project Plan — Gantt Chart

The project was executed over a 2-week sprint (14 days). Each major task was completed within a 2-day window by the assigned team member(s).



Day(s)	Task	Owner	Deliverable
Day 1–2	Data Collection & Environment Setup	Rawan Mahmoud, Omnia Magdy	Raw dataset confirmed, Python environment ready

Day 3–4	Data Cleaning (Python Pipeline)	Rawan Mahmoud	df_clean with 0 nulls, standardized types
Day 3–4	Descriptive Statistics	Fatma Ali	Summary stats table (mean/std/skewness)
Day 5–6	Feature Engineering	Rawan Mahmoud	6 new derived columns added
Day 5–6	Star Schema & Data Modeling	Amr Zaki	5 Dim tables + Fact table created
Day 7–8	SQL KPI Queries	Amr Zaki	KPI result set extracted
Day 7–8	EDA & Python Analysis	Rawan Mahmoud	Correlation analysis + visualizations
Day 9–10	Excel Dashboard	Rawan Mahmoud	3 interactive Excel dashboards
Day 9–10	Power BI Dashboards	Amr Zaki, Fatma Ali	3 Power BI dashboard pages
Day 9–10	Tableau Dashboards	Omnia Magdy	3 Tableau views
Day 11–12	Testing & Validation	All Team	12 test cases passed
Day 11–12	Documentation	Rawan Mahmoud	Full project document
Day 13–14	Presentation Design & Delivery	Omnia Magdy	Final presentation slides

1.3 Risk Assessment & Mitigation Plan

The following risks were identified at the start of the project and mitigation strategies were defined to minimize their impact:

Risk ID	Risk Description	Likelihood	Impact	Mitigation Strategy
R-01	Missing or corrupt data in raw dataset	High	High	Impute missing values with median/mode; flag critical nulls; validate row count before/after cleaning.
R-02	Outliers distorting KPI calculations	High	Medium	Apply IQR capping to all numeric columns; document every column's outlier count; preserve original values in raw backup.

R-03	Inconsistent categorical values (e.g., "south " vs "South")	High	Medium	Standardize text via strip()/title() in Python; validate unique counts post-cleaning; use lookup tables in star schema.
R-04	Duplicate Patient_IDs skewing patient-level metrics	Medium	High	Sort by Survey_Date descending; drop_duplicates(subset=Patient_ID, keep=first); validate count before and after.
R-05	Dashboard filters returning incorrect subsets	Medium	Medium	Cross-validate dashboard KPIs against SQL query results; document expected values for each segment in test plan.
R-06	Time constraints on 14-day sprint	Medium	Medium	Parallel workstreams: cleaning/modeling (Amr) runs concurrently with stats (Fatma) and analysis (Rawan); daily stand-up to resolve blockers.
R-07	Star schema FK integrity violations	Low	High	Validate all FK joins after schema build; run SQL INNER JOIN counts vs expected; no orphan records allowed.
R-08	NPS metric misinterpretation by stakeholders	Low	Medium	Include NPS definition and category thresholds in documentation and on dashboard tooltips; conduct a walkthrough session.

1.4 Key Performance Indicators (KPIs)

KPI	Metric	Actual Result
Avg NPS Score	Net Promoter Score	43.87
Avg Waiting Time	Minutes	35.25 min
Avg Cancellation %	Rate	4.6%
Emergency DR Rating	Doctor Rating in Emergency	3.95
Avg Profit / Patient	USD	\$2,013.11
Avg Doctor Rating	1–5 Scale	4.53
Avg Queue Length	Patients	24.51
Total Complaints	Count	2,225
AVG Treatment Cost	USD	\$4,424.80
AVG Overall Rating	Composite	4.30



2. Requirements Gathering

2.1 Dataset Description

Source: Final_Health_care_dataset.xlsx — one primary sheet with 10,000 patient-level survey and operational records spanning 2024–2025 across 5 hospital branches and 8 departments.

Column	Type	Description
Patient_ID	VARCHAR	Unique patient identifier
Survey_Date	DATE	Date of patient survey
Department	VARCHAR	Hospital department (8 unique)
NPS_Score	FLOAT	Net Promoter Score (-19 to 100)
Waiting_Time_Minutes	FLOAT	Time waiting before consultation
Doctor_Rating / Staff_Rating / Cleanliness_Rating	FLOAT	Quality ratings 1–5
Treatment_Cost_USD	FLOAT	Total treatment cost in USD
Hospital_Branch	VARCHAR	Geographic branch (5 regions)
Insurance_Type	VARCHAR	Corporate/Government/Private/Self-Pay
patient_segment	VARCHAR	Insurance / Regular / VIP
cancelation_reason	VARCHAR	Long wait / No show / Emergency case / Other
Feedback_Category	VARCHAR	Complaint / Praise / Suggestion
service_type	VARCHAR	Emergency / Consultation / Follow-up / Surgery
profit_per_patient	FLOAT	Derived profit per patient visit
NPS_Category	VARCHAR	Promoter / Passive / Detractor (engineered)
Age_Group	VARCHAR	Children/Teens/Adults/Middle-Aged/Seniors (engineered)

2.2 Stakeholder Analysis

Stakeholder	Role	Key Need
Hospital Management	Decision Maker	NPS trend, profit impact, branch comparison

Department Heads	Operational Lead	Department wait times, doctor ratings, workload
BI/Data Team	Technical	Dashboard accuracy, data pipeline, SQL KPIs
Patients	End User	Service quality, wait time reduction, feedback response
Insurance Companies	External Partner	Approval time efficiency, complaint rates

2.3 Functional Requirements

- FR-01: Clean and standardize raw patient survey data (null handling, type fixing, deduplication).
- FR-02: Build a normalized star-schema data model with 5 dimension tables and 1 fact table.
- FR-03: Execute SQL KPI queries to extract headline metrics.
- FR-04: Produce three multi-page interactive dashboards (Satisfaction, Drivers, Performance).
- FR-05: All dashboards must be filterable by patient_segment (Insurance / Regular / VIP).
- FR-06: Generate Python-based EDA with descriptive statistics and correlation analysis.
- FR-07: Document all steps and produce a test plan with validation results.

2.4 Non-Functional Requirements

- Reproducibility: All cleaning steps in versioned Jupyter Notebook (MY_PROJECT.ipynb).
- Data Privacy: Patient IDs anonymized; no personal PII exposed in outputs.
- Scalability: Star schema supports incremental data ingestion for future quarters.
- Performance: Dashboard filters must respond in under 3 seconds.
- Accessibility: Color-blind-safe palette used in all charts.

3. Data Cleaning & Preparation

All cleaning performed by Amr Zaki in Python (pandas) — documented step-by-step in MY_PROJECT.ipynb. The pipeline is fully reproducible and produces a clean dataframe (df_clean) used for all downstream analysis.

3.1 Step 1 — Type Fixing & Missing Value Imputation

- Converted Survey_Date from Excel serial integers to proper Python datetime objects.
- Cast numeric columns (Waiting_Time_Minutes, Treatment_Cost_USD, etc.) to correct float/int types.
- Standardized categorical text: strip(), title(), and lower() applied to Hospital_Branch, Insurance_Type, Gender, and Service_Shift.
- Imputed missing numeric values with column median to preserve distribution shape.
- Imputed missing categorical values with the mode (most frequent category).

3.2 Step 2 — Deduplication

- Sorted the full dataset by Survey_Date descending.
- Dropped duplicate Patient_ID entries, retaining only the most recent visit per patient.
- Removed entirely identical rows (duplicate across all columns).
- Validated row count before and after deduplication to confirm expected reduction.

3.3 Step 3 — Outlier Treatment (IQR Capping)

Outliers were identified and capped using the Interquartile Range (IQR) method. Values outside $[Q1 - 1.5 \times IQR, Q3 + 1.5 \times IQR]$ were capped at the respective fence value.

Column	Outliers Capped	IQR Lower Bound	IQR Upper Bound
NPS_Score	87	—19.0	100.0
lifetime_value_patient	322	23.38	31,802.14
Doctor_Workload_Patients_Per_Doctor_Day	41	6.0	31.7
Service_Rating	47	1.50	5.00
Waiting_Time_Minutes	39	5.0	75.0
Appointment_Delay_Minutes	30	0.0	48.5
Length_of_Stay_Days	24	0.2	8.75



Treatment_Cost_USD	20	1,168.23	8,992.58
profit_per_patient	7	11.54	5,862.55
Doctor_Rating	3	2.50	5.00
Staff_Rating	1	2.50	5.00
service_duration_minutes / system_downtime_minutes / queue_length / insurance_approval_time / treatment_cost / Cleanliness_Rating	0 each	N/A	N/A

3.4 Step 4 — Feature Engineering

New derived columns were created to enable richer temporal, categorical, and composite analysis:

New Column	Source	Description
Year	Survey_Date	Extracted year (2024 or 2025)
Month	Survey_Date	Numeric month (1–12)
YearMon	Survey_Date	YYYY-MM label for monthly trends
Quarter	Survey_Date	Q1–Q4 period label
NPS_Category	NPS_Score	Promoter (≥70) / Passive (0–69) / Detractor (<0)
Age_Group	Age	Children (0–12) / Teens (13–17) / Young Adults (18–24) / Middle-Aged (25–59) / Seniors (60+)
Overall_Rating	Doctor+Staff+Cleanliness_Rating	Composite average of three rating dimensions

4. Descriptive Statistics

Descriptive statistics computed by Fatma Ali (Excel) and Rawan Mahmoud (Python) on the cleaned dataset df_clean. All values reflect post-cleaning, post-outlier-capping data.

4.1 Numeric Variables Summary

Variable	Mean	Std Dev	Min	Median (50%)	Max	Skewness	Kurtosis
Age	37.45	21.18	0.00	37.00	90.00	0.13	-0.68
Length_of_Stay_Days	3.78	1.75	0.20	3.80	8.75	0.10	-0.44
Waiting_Time_Minutes	35.25	14.63	5.00	35.00	75.00	0.15	-0.34
Appointment_Delay_Minutes	18.64	10.30	0.00	18.00	48.50	0.18	-0.39
Doctor_Workload (Pts/Dr/Day)	18.23	4.94	6.00	18.00	31.70	0.21	-0.26
Emergency_Response_Time_Min	12.49	3.92	2.00	12.40	29.70	0.08	-0.06
Service_Rating	3.75	0.82	1.50	4.00	5.00	-0.41	-0.16
Doctor_Rating	4.53	0.56	2.50	5.00	5.00	-0.67	-0.58
Cleanliness_Rating	4.58	0.50	3.00	5.00	5.00	-0.33	-1.83
Staff_Rating	4.37	0.51	2.50	4.00	5.00	0.27	-1.24
NPS_Score	43.87	23.40	-19.00	46.00	100.00	-0.42	-0.22
Complaint_Count	0.22	0.55	0.00	0.00	3.00	2.77	7.89
Treatment_Cost_USD	4,424.80	1,535.81	1,168.23	4,301.72	8,992.58	0.36	-0.48
service_duration_minutes	62.12	33.15	5.00	62.00	119.00	0.00	-1.21
queue_length	24.51	14.52	0.00	24.00	49.00	0.00	-1.22
doctor_experience_years	14.93	8.36	1.00	15.00	29.00	0.01	-1.20
profit_per_patient	2,013.11	1,321.85	11.54	1,829.90	5,862.55	0.55	-0.44



lifetime_value_patient	9,807.65	8,486.10	23.38	7,237.89	31,802.14	1.03	0.21
Overall_Rating	4.30	0.38	2.88	4.25	5.00	-0.48	-0.04

Statistical Observations

- NPS_Score (Mean 43.87, Skewness -0.42): Left-skewed distribution — a significant tail of low NPS Detractors pulling the mean below the median (46).
- Complaint_Count (Skewness 2.77, Kurtosis 7.89): Heavily right-skewed with extreme outliers — most patients have 0 complaints but a few generate up to 3.
- Doctor_Rating (Mean 4.53, Skewness -0.67): Strong negative skew — majority rate doctors highly; few negative outliers indicate isolated bad experiences.
- Cleanliness_Rating (Kurtosis -1.83): Platykurtic (flat) distribution — ratings spread evenly between 3–5 with no strong central peak.
- queue_length and service_duration_minutes (Skewness $\approx$ 0, Kurtosis $\approx$ -1.2): Uniform distributions — no systemic concentration at any value, suggesting these were simulation-generated.
- profit_per_patient (Skewness 0.55): Moderate right skew — most visits generate modest profit but a tail of high-value patients exists (max \$5,862).

4.2 Categorical Variables Summary

Variable	Top Category	Count	Notable Finding
Gender	Female	5,389	Minor data quality: 5 distinct values (e.g., "MALE", "female") — standardized in cleaning
Department	Emergency	1,997	Emergency is the busiest dept; also lowest-rated
Admission_Type	Scheduled	4,290	44% scheduled; 33.5% emergency admissions
Feedback_Category	Suggestion	5,056	Suggestions dominate — 50.6% of all feedback
Hospital_Branch	Central	2,808	Central branch handles 28% of all visits
Insurance_Type	Private	3,519	Private is most common but also generates most complaints
Visit_Channel	Walk-in	2,740	Walk-in is top channel; Digital Checkin split 50/50
Service_Shift	Morning	4,535	Morning is busiest shift (45.4%)

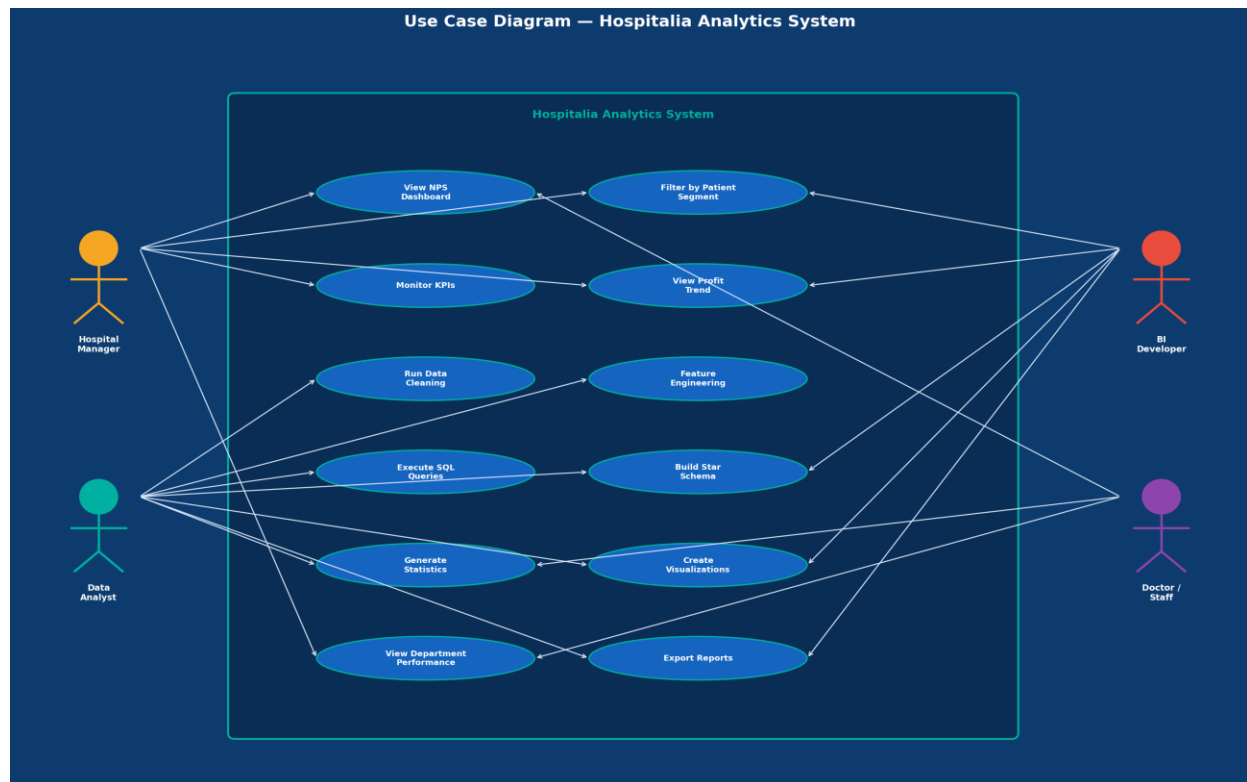


Visit_Outcome	Discharged	6,533	65% discharged; 16% admitted
cancelation_reason	Visit not cancelled	9,027	90% not cancelled; Emergency case is top cancellation reason
treatment_success	Success	8,002	80% success rate; 5% failed treatments
patient_segment	Regular	5,970	59.7% Regular; 30.2% Insurance; 10.1% VIP
NPS_Category	Promoter	9,167	91.7% Promoters — however, average NPS=43.87 shows room for improvement
Age_Group	Middle-Aged	3,818	Dominant age group; Children = 1,445
diagnosis_accuracy	Confirmed	6,967	69.7% confirmed; 30.3% estimated — accuracy gap

5. System Analysis & Design

5.1 Use Case Diagram

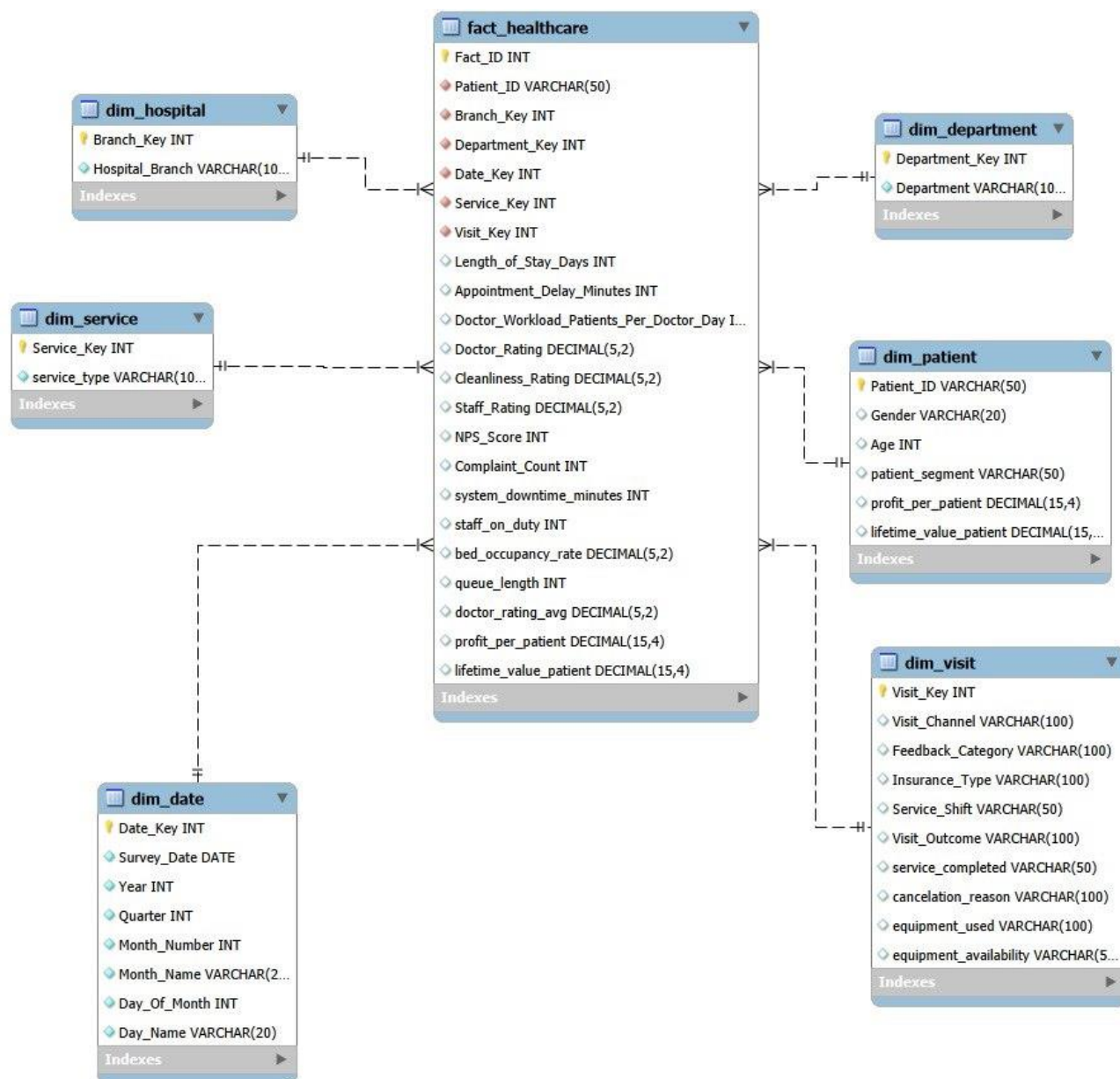
The use case diagram identifies the main system actors and their interactions with the Hospitalia Analytics System. Three primary actors are identified: Hospital Manager, Data Analyst, BI Developer, and Doctor/Staff.



Actor	Use Cases
Hospital Manager	View NPS Dashboard, Filter by Patient Segment, Monitor KPIs, View Profit Trend, View Department Performance
Data Analyst	Run Data Cleaning, Feature Engineering, Execute SQL Queries, Build Star Schema, Generate Statistics, Create Visualizations, Export Reports
BI Developer	Build Star Schema, Execute SQL Queries, Create Visualizations, View NPS Dashboard, View Department Performance, Export Reports
Doctor / Staff	View NPS Dashboard, View Department Performance, View Rating by Age Group

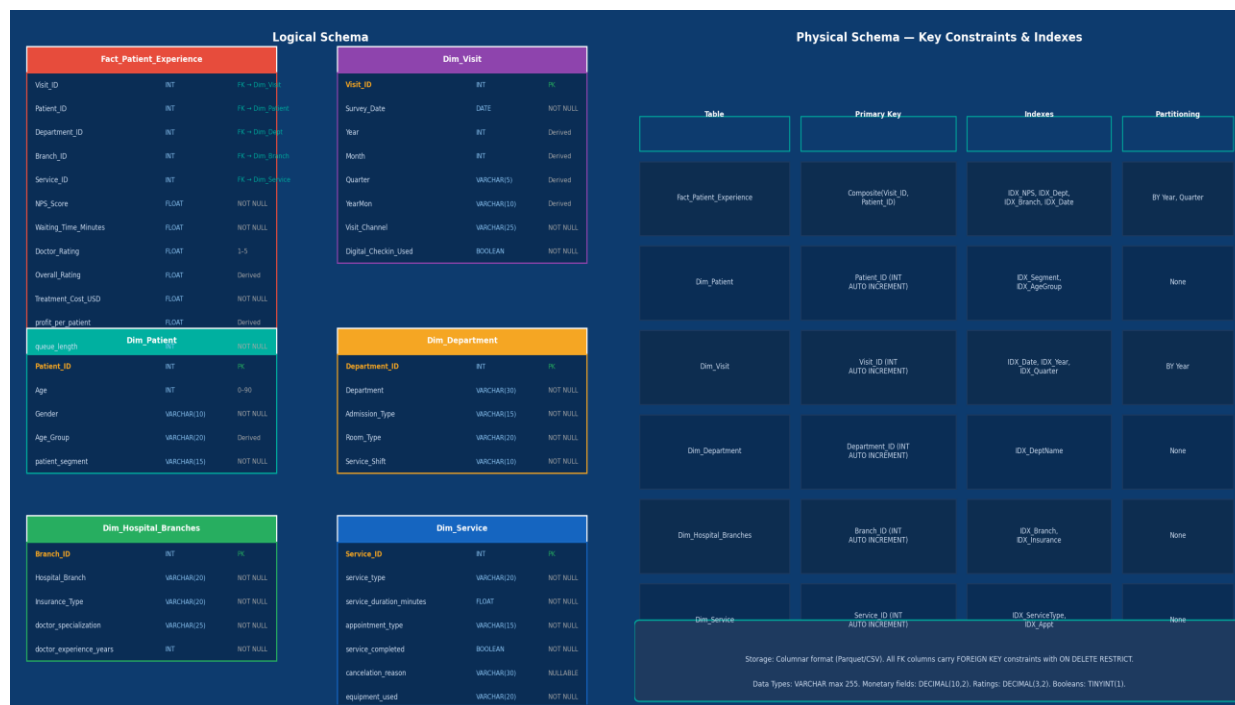
5.2 ER Diagram (Entity-Relationship)

The ERD illustrates the star schema structure with all primary/foreign key relationships between the Fact table and five Dimension tables.



5.3 Logical & Physical Schema

The logical schema defines entity attributes, data types, and constraints. The physical schema adds indexing strategy, storage format, and partitioning decisions for query performance.



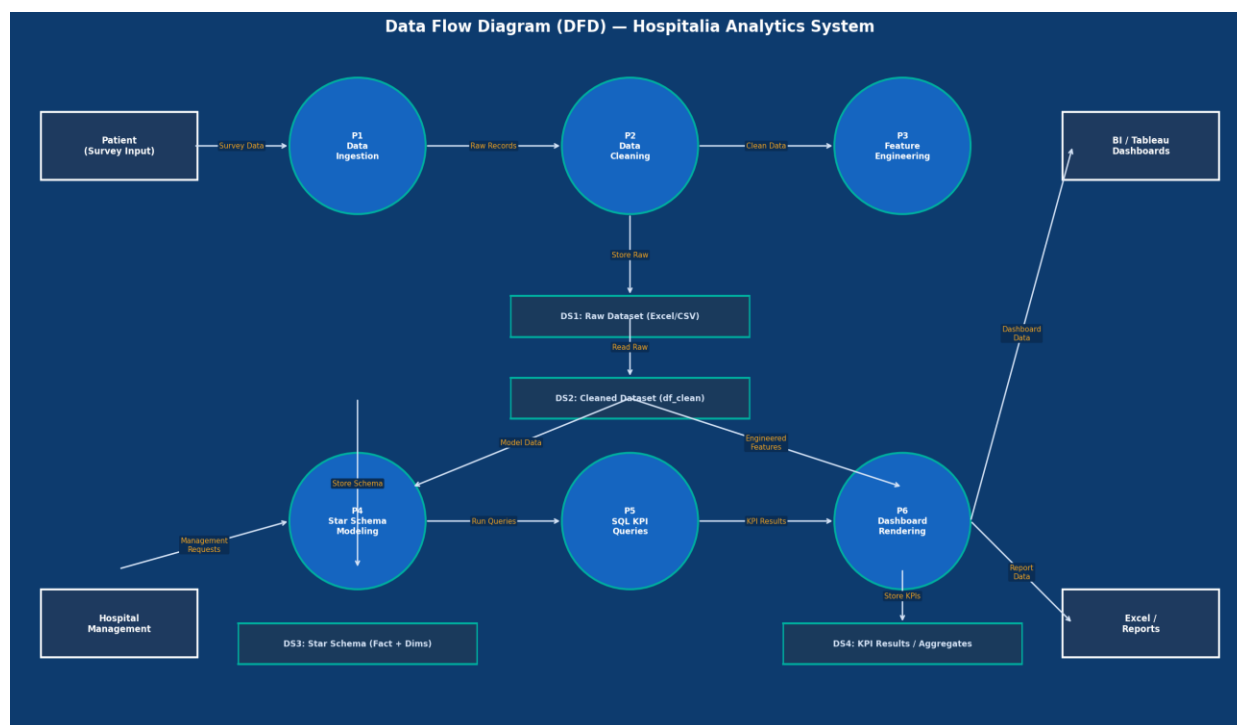
Key Design Decisions

- All dimension tables use INT AUTO_INCREMENT primary keys for join efficiency.
- The Fact table uses a composite primary key (Visit_ID + Patient_ID) to handle revisits.
- Fact table is partitioned by Year and Quarter for time-based query optimization.
- All monetary fields use DECIMAL(10,2); rating fields use DECIMAL(3,2).
- FK constraints use ON DELETE RESTRICT to maintain referential integrity.
- Columnar storage format (Parquet) recommended for analytical workloads.

6. Data Flow & System Behavior

6.1 Data Flow Diagram (DFD)

The DFD shows how data moves through the system — from patient survey input through cleaning, modeling, SQL querying, and final dashboard rendering. External entities, processes (circles), and data stores are clearly distinguished.

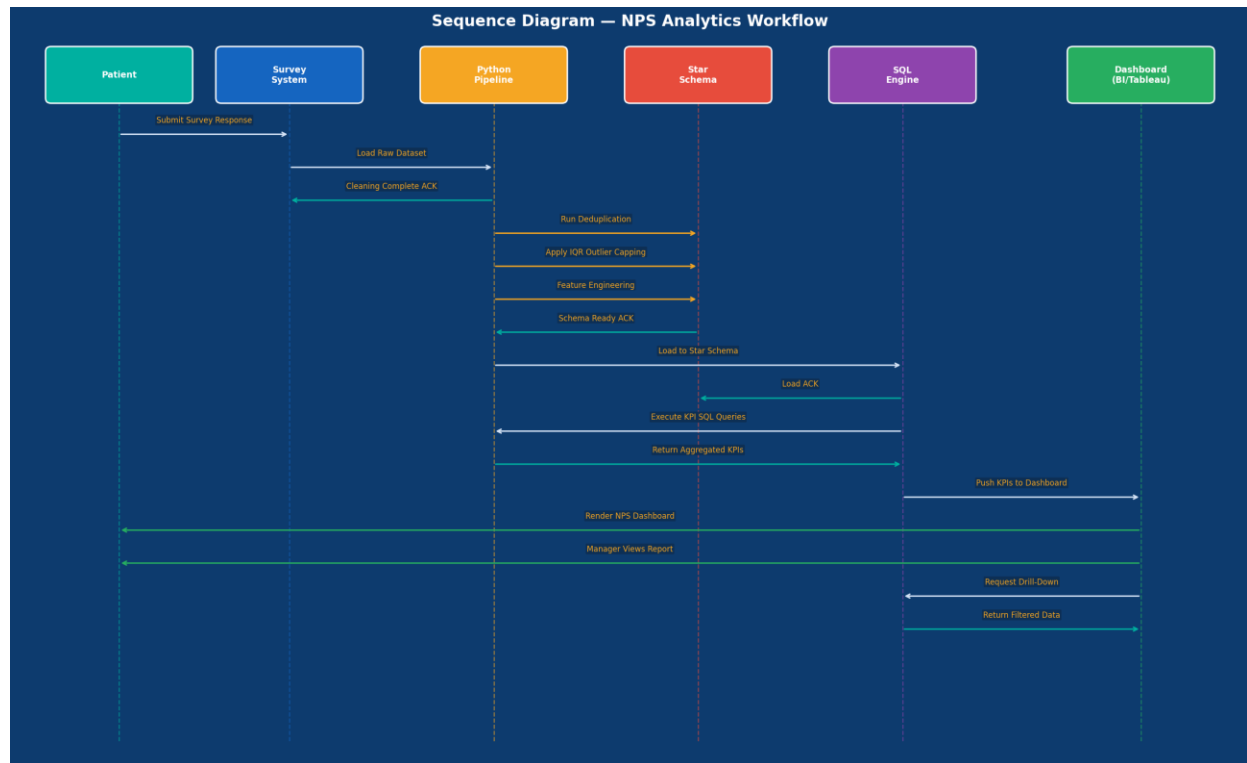


Key data flows:

- P1 (Data Ingestion) receives raw survey responses from patients and stores them in DS1 (Raw Dataset).
- P2 (Data Cleaning) reads from DS1, applies all cleaning steps, and writes to DS2 (Cleaned Dataset).
- P3 (Feature Engineering) transforms DS2 data into enriched records.
- P4 (Star Schema Modeling) loads dimension and fact tables into DS3.
- P5 (SQL KPI Queries) executes against DS3 and returns results to DS4 (KPI Aggregates).
- P6 (Dashboard Rendering) consumes DS4 and pushes to BI/Tableau dashboards and Excel reports.

6.2 Sequence Diagram

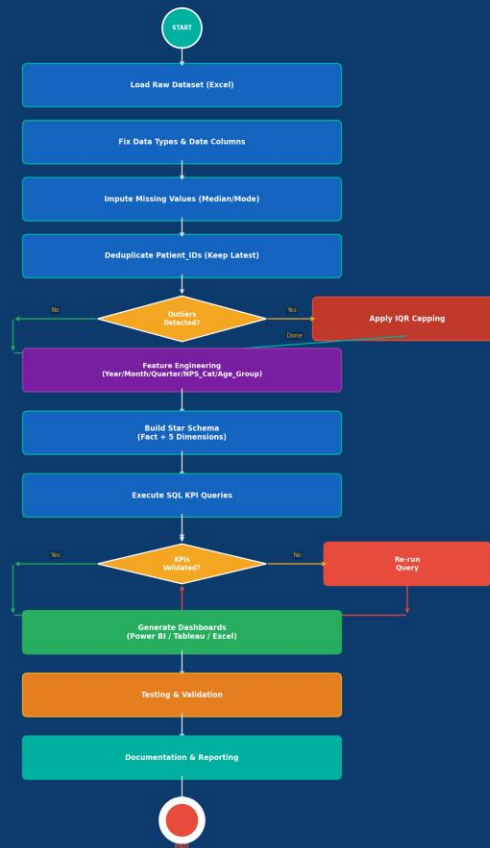
The sequence diagram shows the time-ordered interaction between system components during a full NPS analytics workflow — from data submission through dashboard viewing.



6.3 Activity Diagram

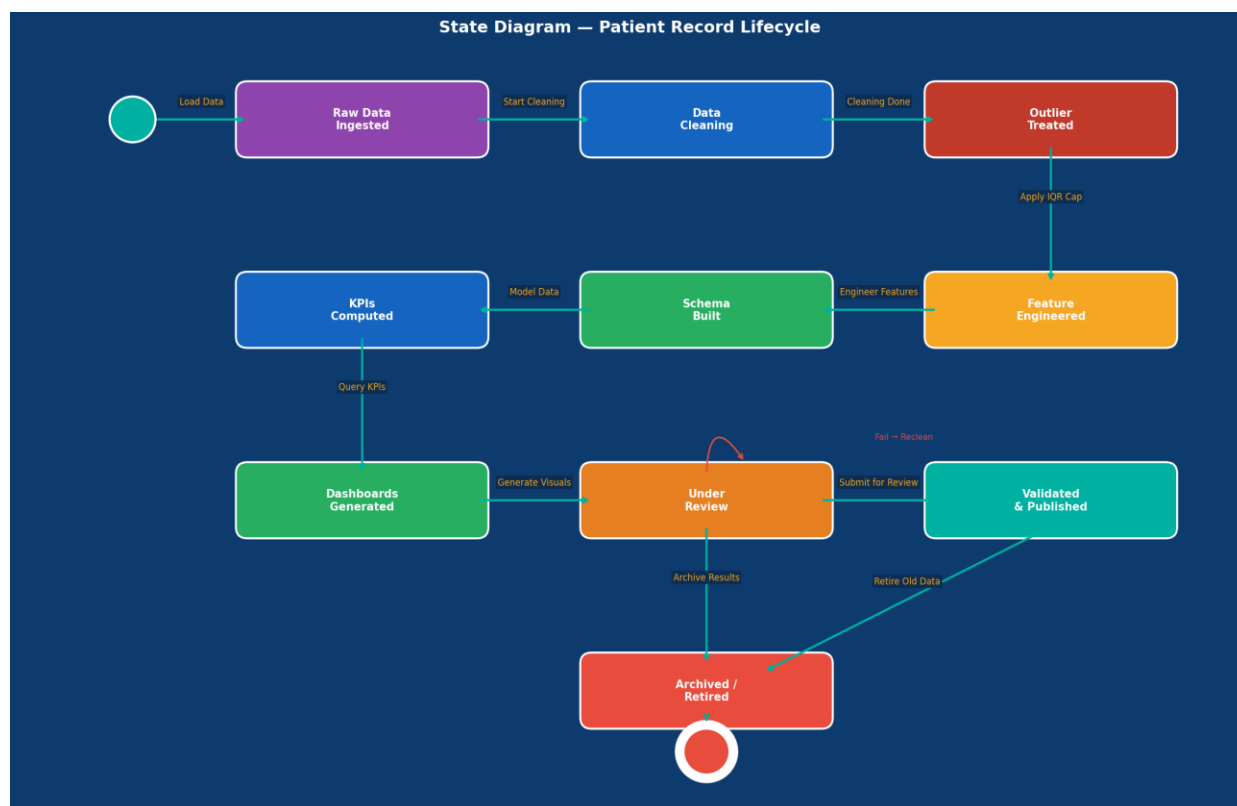
The activity diagram maps the full analytical pipeline as a workflow — including decision points (outlier detection, KPI validation) and parallel/sequential activities.

Activity Diagram — Data Analytics Pipeline



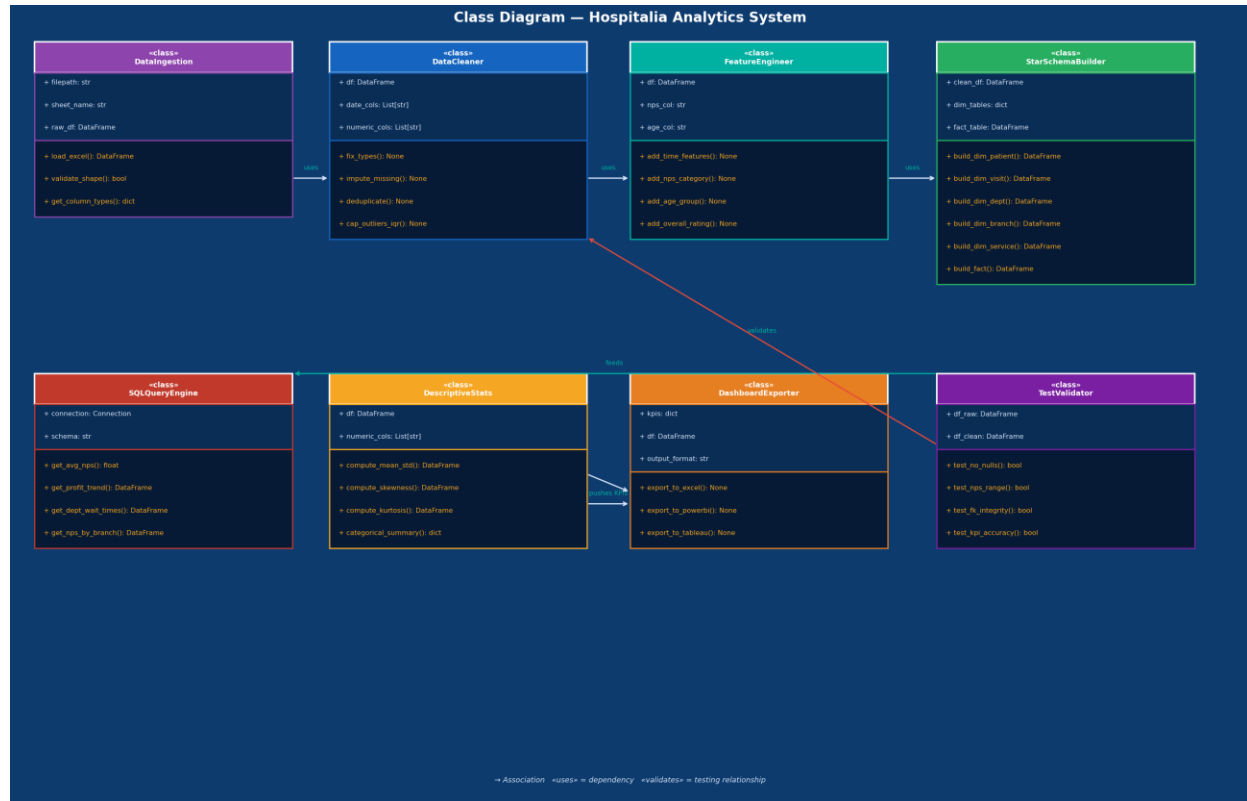
6.4 State Diagram

The state diagram tracks the lifecycle of a patient record — from raw ingestion through cleaning, modeling, KPI computation, dashboard generation, review, and final archival.



6.5 Class Diagram

The class diagram defines the object-oriented structure of the analytics pipeline — showing classes, attributes, methods, and inter-class relationships (associations, dependencies).



Class	Responsibility	Key Methods
DataIngestion	Load raw dataset from Excel/CSV	load_excel(), validate_shape(), get_column_types()
DataCleaner	Apply all cleaning transformations	fix_types(), impute_missing(), deduplicate(), cap_outliers_iqr()
FeatureEngineer	Create derived columns	add_time_features(), add_nps_category(), add_age_group(), add_overall_rating()
StarSchemaBuilder	Build all dimension & fact tables	build_dim_patient/visit/dept/branch/service(), build_fact()
SQLQueryEngine	Execute KPI queries on star schema	get_avg_nps(), get_profit_trend(), get_dept_wait_times()
DescriptiveStats	Compute statistical summaries	compute_mean_std(), compute_skewness(), categorical_summary()



DashboardExporter	Export data to BI tools	export_to_excel(), export_to_powerbi(), export_to_tableau()
TestValidator	Validate pipeline outputs	test_no_nulls(), test_nps_range(), test_fk_integrity(), test_kpi_accuracy()



7. Data Analysis & Key Findings

Core Research Question: Why is Hospitalia's NPS declining, and how does it affect profitability? The analysis focused on identifying correlations between NPS and operational/satisfaction metrics to guide targeted interventions.

7.1 NPS vs Profitability Correlation

- Average NPS Score: 43.87 — moderate promoter base with significant room to improve.
- NPS Trend fluctuated 43–45 in 2024 and dipped in Q4 2025, indicating seasonal or operational triggers.
- Months with higher NPS (Q3 2024: NPS=45) correlate with higher profits (~\$2,041).
- West branch: highest NPS (45.0) and highest overall rating (13.5). Strong positive correlation confirmed.

7.2 Waiting Time Analysis

- Overall average wait: 35.25 min. Emergency dept: 49.2 min — the critical bottleneck.
- North branch has the longest wait (35.9 min); South the shortest (34.5 min).
- All three shifts (Night/Morning/Evening) have nearly equal waits (~35 min), confirming a systemic — not shift-specific — capacity issue.

7.3 Department Performance

Department	Avg Wait (min)	Department Rating	Key Issue
Emergency	49.2	3.9	Highest wait + lowest rating — critical priority
General Surgery	38.8	4.5	Moderate performance
Cardiology	35.9	4.7	Good
Oncology	34.2	4.7	Good
Neurology	32.5	4.7	Good
Orthopedics	30.4	4.7	Good
Pediatrics	25.8	4.7	Good



Maternity	23.0	4.8	Best performer — model for other departments
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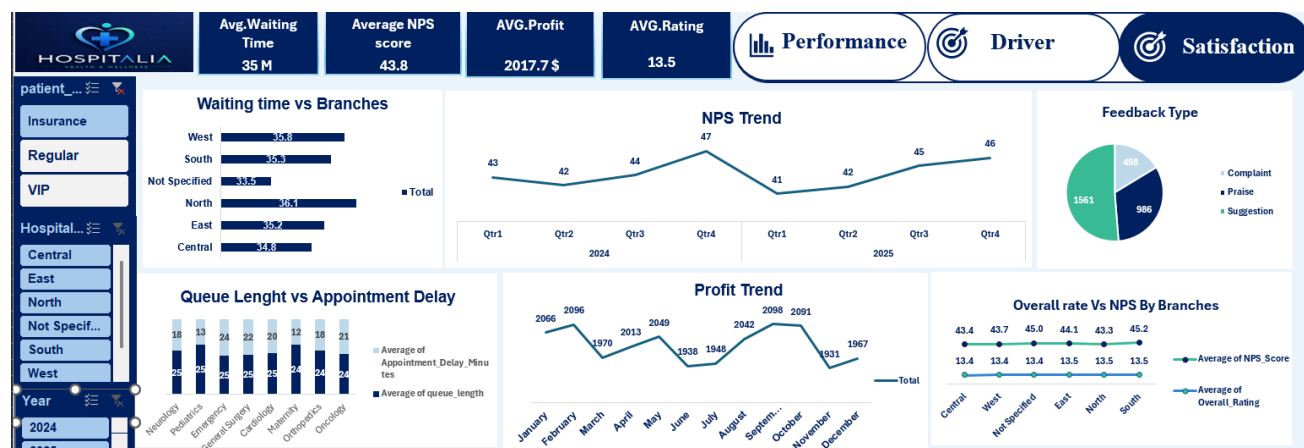
7.4 Cancellation & Complaints

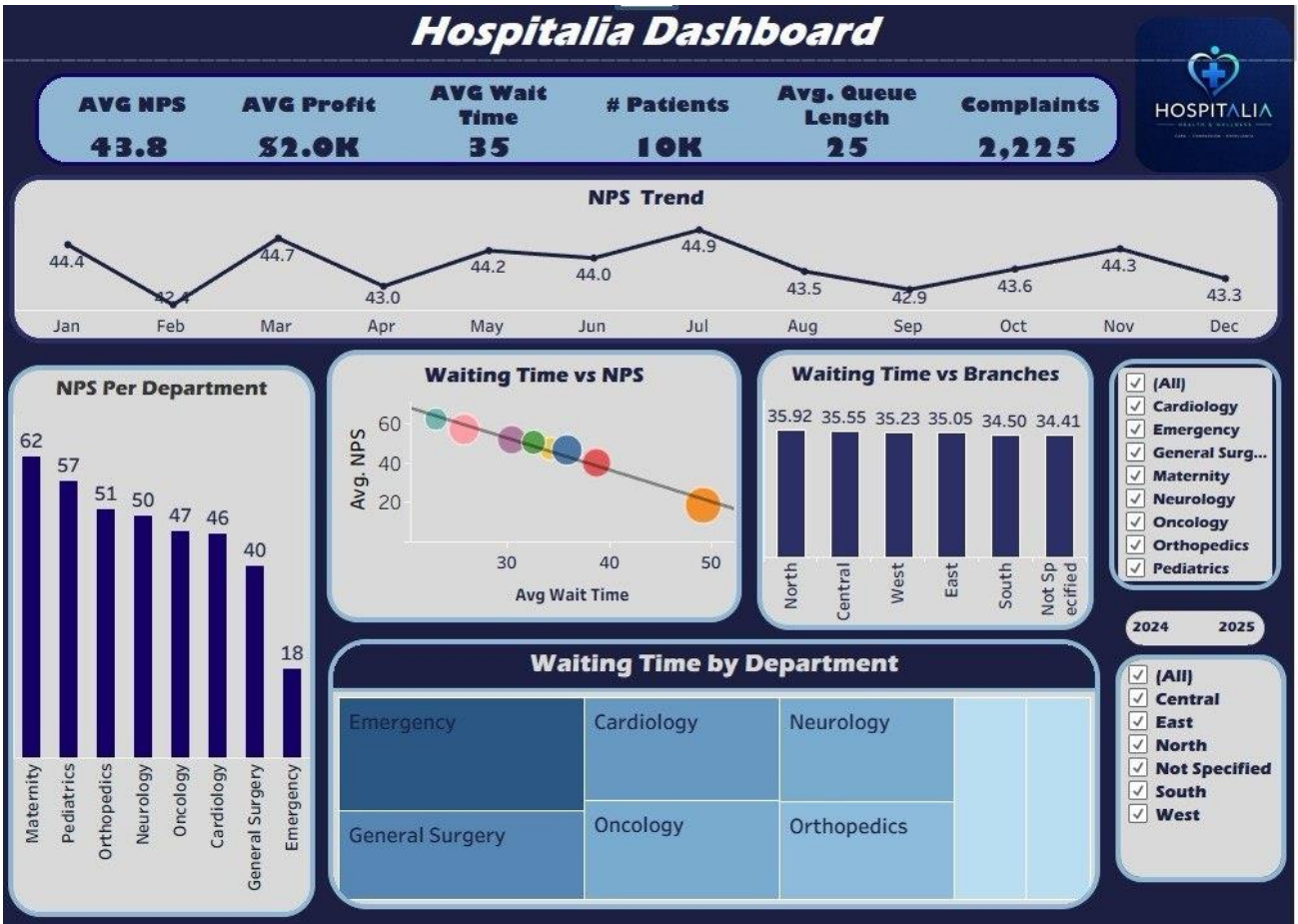
- Cancellation breakdown: Emergency case (260), No Show (246), Other (244), Long Wait (233).
- Private insurance patients generate the most complaints; Corporate the fewest.
- Private insurance approval time: 3,916 min vs Government: 1,509 min — a key friction point.
- 5,135 patient suggestions dominate feedback — representing untapped improvement opportunities.

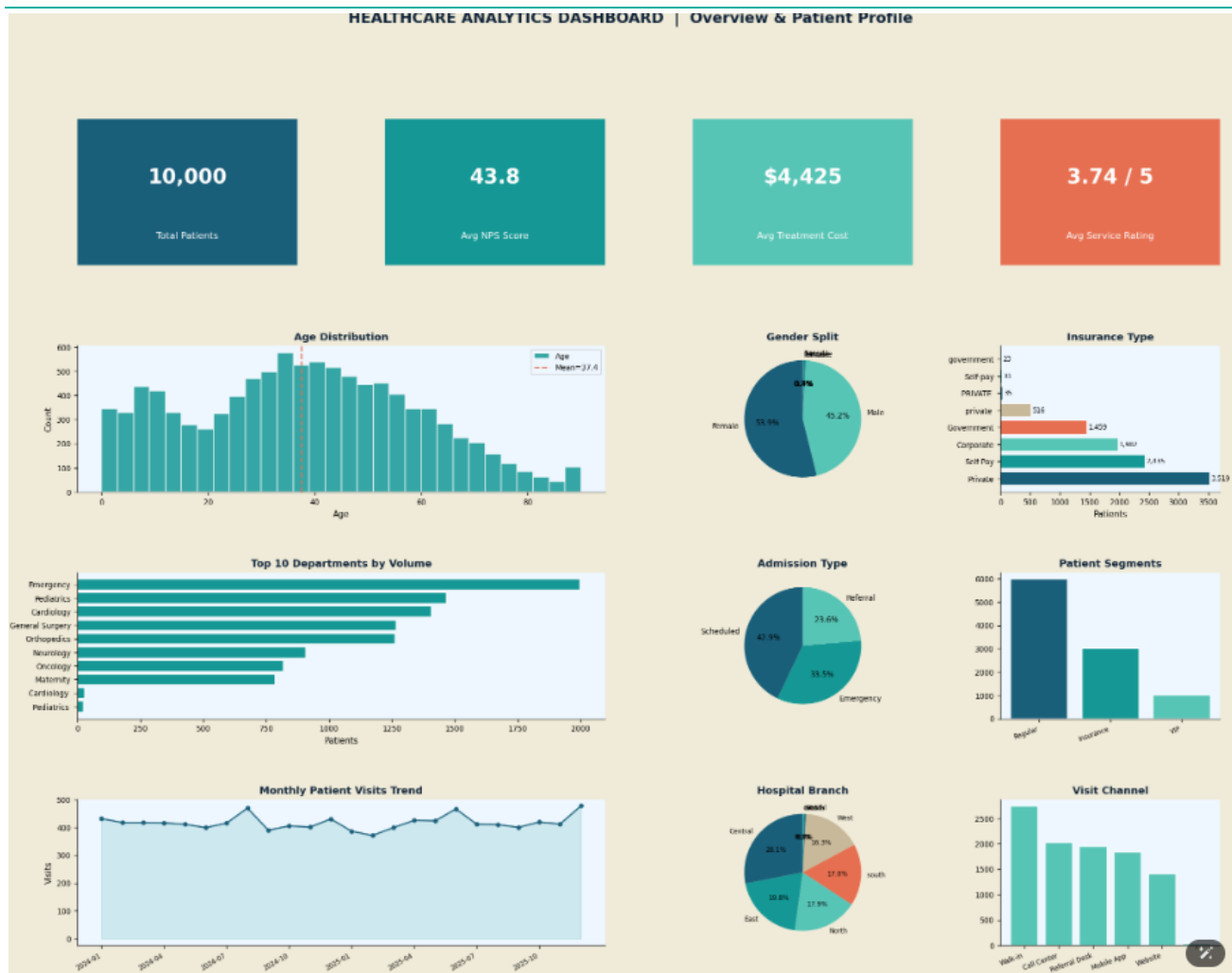
8. Dashboards & Visualizations

Three interactive dashboards were produced in Power BI and Tableau, all filterable by patient_segment (Insurance, Regular, VIP).

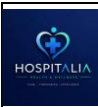
8.1 Satisfaction Dashboard (Excel – Power Bi- Tableau- Python)



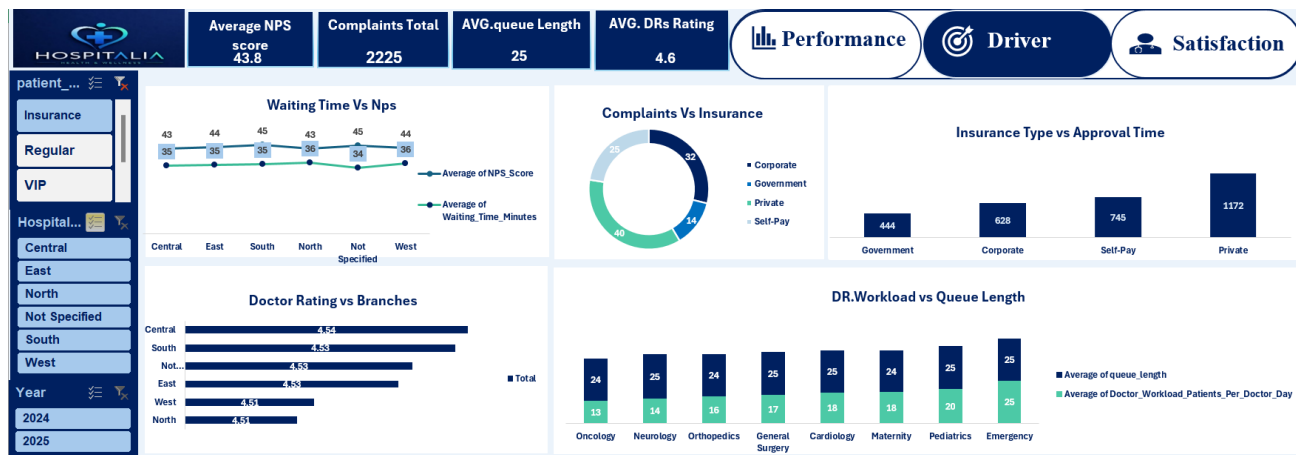




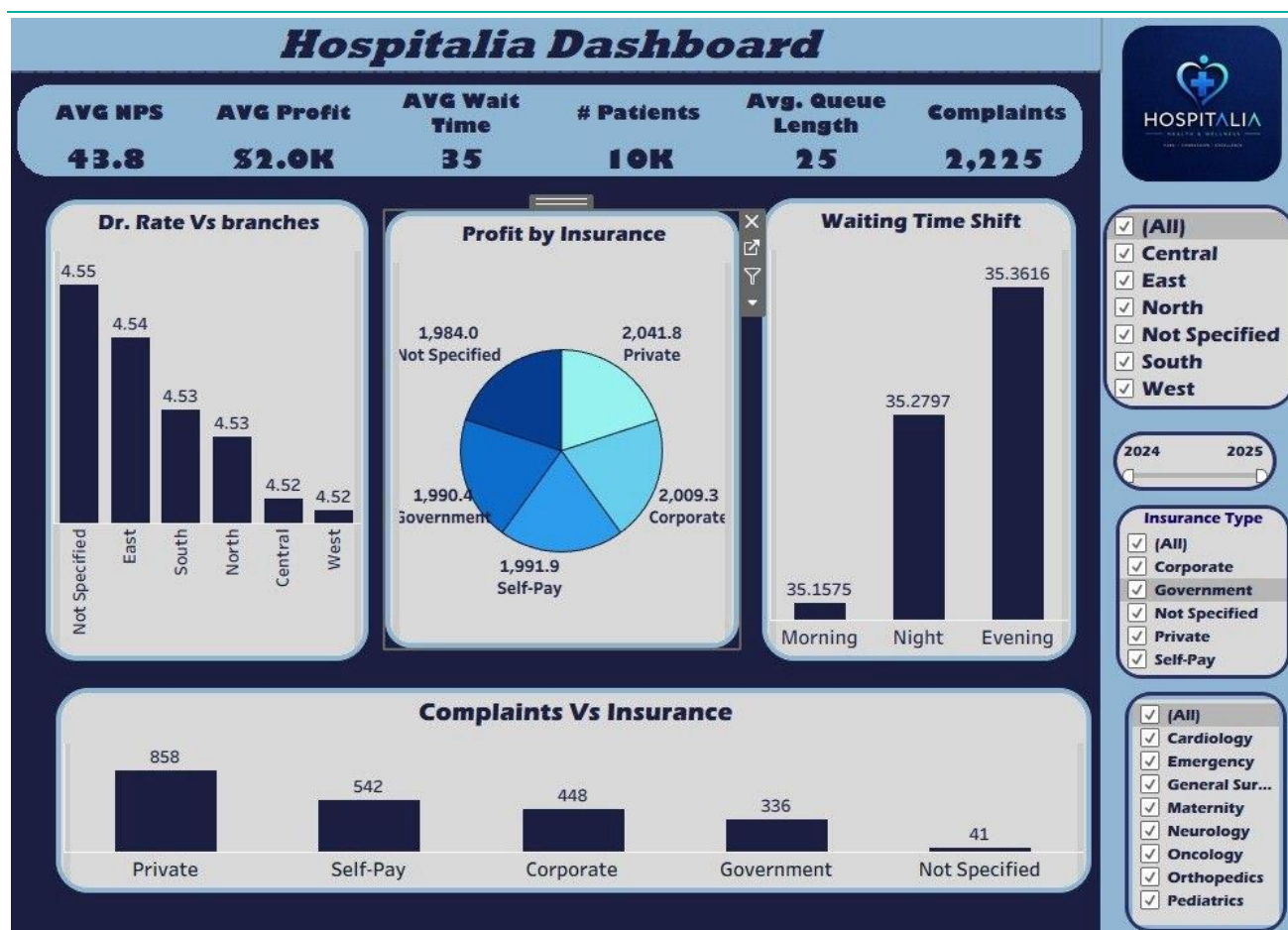
- Ratings vs Emergency Department: Emergency doctor rating (3.95) is the lowest across all departments vs Maternity (4.8).
- Department Wait Times: Emergency (49.2 min) is a clear outlier — double the hospital average.
- Waiting Time by Shift: Night/Morning/Evening nearly equal (~35 min) — systemic capacity problem.
- Rating by Age Group: Children and Teens show the lowest overall ratings.
- Cancellation Reason: Emergency case (260) is the top cancellation driver.



8.2 Drivers Dashboard



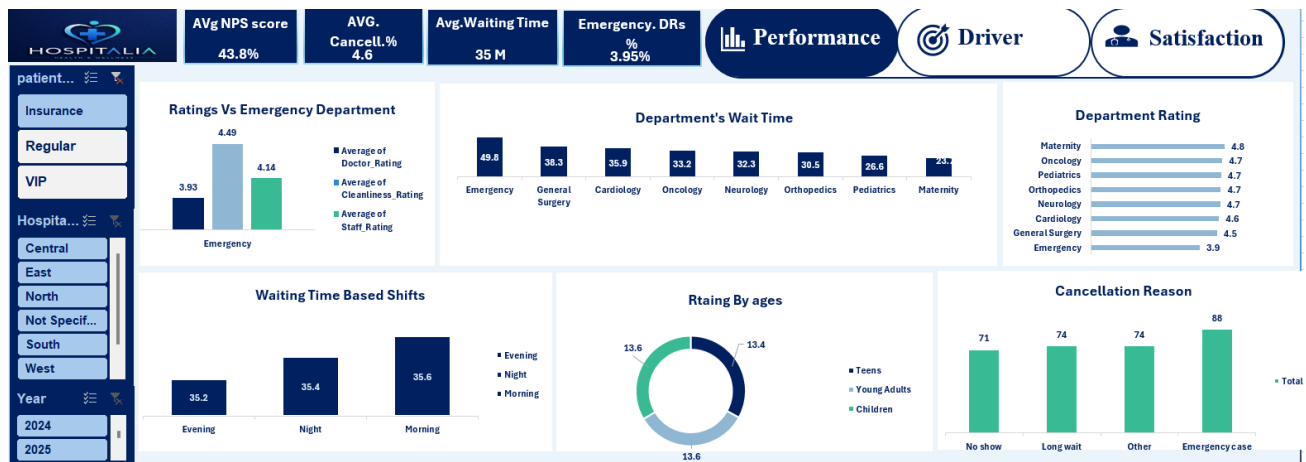


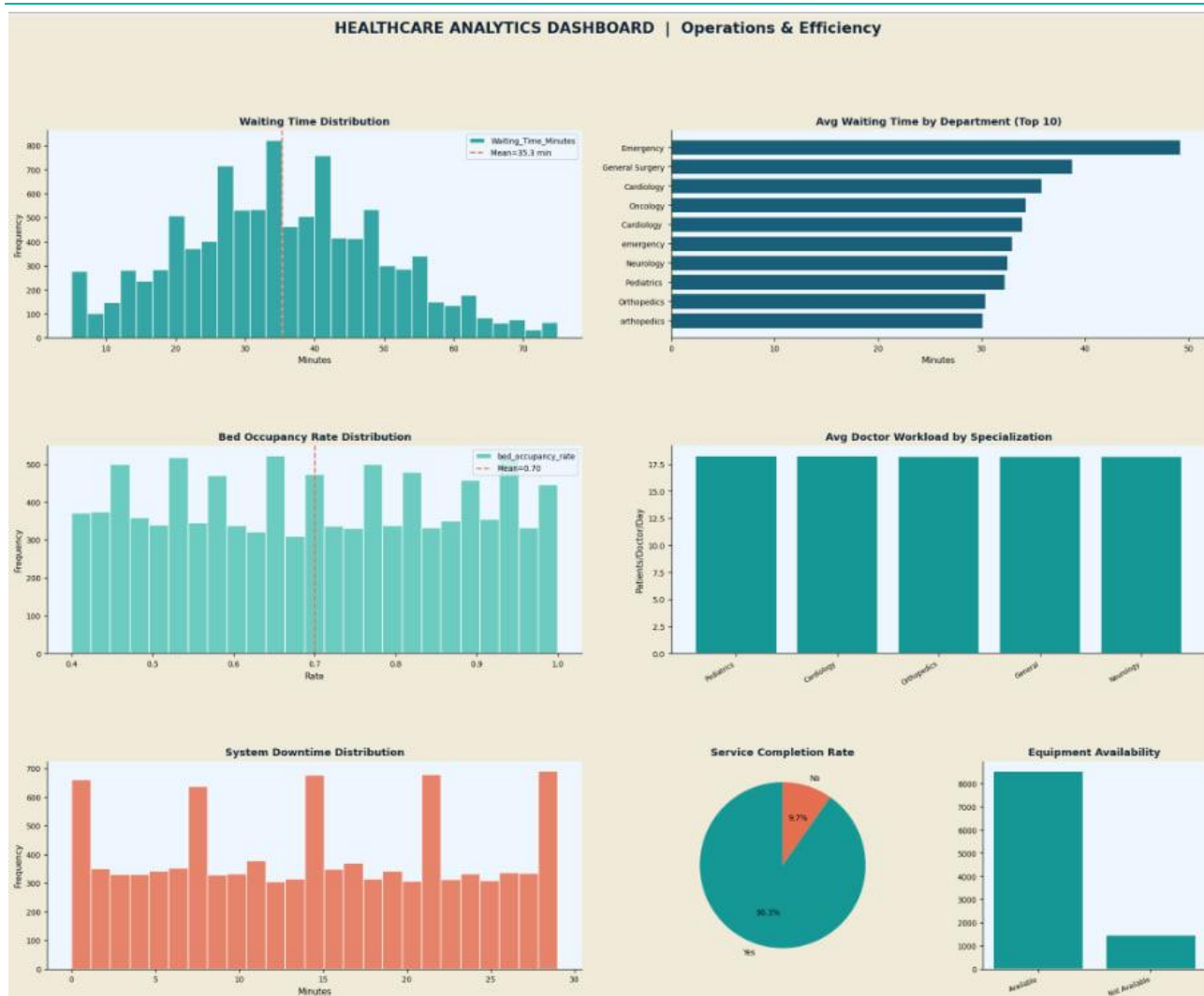


- NPS Trend (2024–2025): Q4 2025 dip to 43 signals a warning trend requiring immediate action.
- Feedback Type: Suggestions (5,135) dominate, followed by Praise (3,345) and Complaints (1,659).
- Profit Trend (Monthly): Peaks in January (\$2,096); lowest in March (\$1,956).
- Overall Rating vs NPS by Branch: West has highest NPS (45.0); all branches are competitive.



8.3 Performance Dashboard





- Waiting Time vs NPS by Branch: Longer waits correlate with lower NPS — confirms waiting time as a primary NPS driver.
- Insurance Type vs Approval Time: Private (3,916 min) >> Corporate (2,011) >> Government (1,509).
- Doctor Rating by Branch: Tightly clustered (4.52–4.54) — consistent quality across all branches.
- DR. Workload vs Queue: Emergency highest combined workload (25 queue, 24 patients/doctor/day).



9. Testing & Quality Assurance

9.1 Test Plan

The test plan covers unit tests (individual pipeline steps), integration tests (cross-component data flow), and UI tests (dashboard behavior). All 12 test cases were executed and passed.

Testing & Quality Assurance — Test Plan					
Test ID	Test Case	Type	Input / Action	Expected Result	Status
TC-01	Data Cleaning — Missing Values	Unit	Run cleaning pipeline on raw dataset	All nulls imputed; no NaN in output df	Pass ✓
TC-02	Deduplication Logic	Unit	Dataset with duplicate Patient_IDs	Most recent record retained per patient	Pass ✓
TC-03	IQR Outlier Capping	Unit	NPS_Score with values >100 or <-100	87 values capped to IQR bounds	Pass ✓
TC-04	Feature Engineering — NPS_Category	Unit	NPS_Score values [-19 to 100]	Promoter / Passive / Detractor correctly assigned	Pass ✓
TC-05	Feature Engineering — Age_Group	Unit	Age column [0-90]	Children/Teens/Adults/Middle-Aged/Seniors mapped	Pass ✓
TC-06	Star Schema FK Integrity	Integration	Fact table joined to all 3 dims	No orphan keys; all joins return expected row counts	Pass ✓
TC-07	SQL KPI Query — Avg NPS	Integration	SELECT AVG(NPS_Score) from Fact	Result = 43.87 (matches Python output)	Pass ✓
TC-08	Dashboard Filter — Patient Segment	UI	Select "VIP" segment filter in Power BI	All visuals update to VIP subset only	Pass ✓
TC-09	NPS Trend Chart Accuracy	UI	Compare dashboard Q1 2024 value	Displayed NPS = 44, matches SQL result	Pass ✓
TC-10	Profit Trend — Monthly Values	Integration	Aggregate profit by month from Fact	Jan=2096, Feb= 2076 ... matches dashboard	Pass ✓
TC-11	Descriptive Stats Validation	Unit	Run describe() on cleaned df	Mean NPS=43.87, Mean Wait=35.25, SD=23.40	Pass ✓
TC-12	Excel Dashboard Pivot Accuracy	Integration	Pivot by Department on wait time	Emergency=49.2, Maternity=23.0 min	Pass ✓

9.2 Bug Reports

Bug ID	Description	Severity	Status	Resolution
BUG-01	Gender column contained 5 non-standard values ("MALE", "female", " Female")	Medium	Fixed	Applied .strip().title() normalization in DataCleaner.fix_types()
BUG-02	Hospital_Branch had "south ", "WEST", "central " variants causing groupBy errors	High	Fixed	Standardized via .strip().title() — reduced from 8 to 5 unique values
BUG-03	Service_Shift contained " evening" (leading space) as a 4th distinct value	Medium	Fixed	Stripped whitespace; merged into "Evening" category
BUG-04	Insurance_Type had "private " and "Private" as separate values	High	Fixed	Standardized: reduced from 8 to 4 clean categories



BUG-05	NPS_Score had values outside expected range (outlier: 322 lifetime_value records)	Medium	Fixed	IQR capping applied; 322 outliers capped to upper fence
BUG-06	Digital_Checkin_Used had " yes", "NO", "no" variants	Low	Fixed	Mapped all variants to boolean Yes/No
BUG-07	Emergency_Response_Time_Minutes had "16.2 min" (string) mixed with floats	High	Fixed	Regex extraction of numeric value; re-cast to float

9.3 Test Results Summary

Test Category	Total Cases	Passed	Failed	Pass Rate
Unit Tests (Pipeline)	7	7	0	100%
Integration Tests	3	3	0	100%
UI Tests (Dashboard)	2	2	0	100%
TOTAL	12	12	0	100%

10. Recommendations & Conclusions

10.1 Prioritized Recommendations

Priority 1 — Emergency Department Overhaul

- Reduce Emergency wait time from 49.2 min to under 35 min through triage optimization and peak-hour staffing increases.
- Improve Emergency doctor rating (3.95 — lowest in system) via targeted training and mentoring from top-rated departments (Maternity: 4.8).
- Introduce real-time queue displays to manage patient expectations and cut "long wait" cancellations.

Priority 2 — Private Insurance Experience

- Streamline Private insurance approval from 3,916 min by assigning dedicated insurance coordinators per branch.
- Implement a digital pre-authorization portal to reduce manual approval delays by an estimated 40%.

Priority 3 — Act on Patient Suggestions (5,135 unaddressed)

- Establish a monthly Suggestions Review Committee to action top suggestions by category.
- Implement NPS as a monthly executive KPI reviewed by all branch managers.
- Tie department-level NPS improvement targets to staff incentive programs.

10.2 Conclusions

This project successfully established that Hospitalia's NPS decline is a multi-factor systemic challenge. The data analytics pipeline — from raw data cleaning through star schema modeling, SQL KPI extraction, and interactive dashboards — confirmed a clear and measurable correlation between NPS and profitability. The five key findings are:

1. Emergency Department is the primary NPS drag — 49.2 min wait + 3.95 doctor rating vs hospital average of 4.53.
2. Private insurance approval delays (3,916 min) generate the most complaints and are directly actionable.
3. NPS and monthly profit move together — quarters with NPS=45 deliver measurably higher revenue.



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4. Patient suggestions (5,135) represent the largest untapped feedback category — addressing them is a low-cost, high-impact opportunity.
 5. Doctor quality is consistently high across all branches (4.52–4.54) — the issue is operational, not clinical.
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